

CallRank

CallRank: Isolating What Changed in Earnings-Call Q&A for Sector Ranking and Cost-Adjusted Alpha

Comparing each company's earnings-call Q&A with its own past calls ranks which sector ETFs will beat SPY. In a 24-month out-of-sample test, buying the top-ranked sector each month returned 42.1% a year after costs, against 24.0% for SPY.

.158

monthly sector-rank IC

raw displacement; Newey–West $p = .007$

42.1% vs 24.0%

annual net return, Top 1 vs SPY

unit direction; 5 bp one-way costs

2.05 vs 1.17

Sharpe ratio, Top 1 vs SPY

max drawdown 9.9% vs 18.8%

Juyeong Lee · 4th author

Accepted at ICAIF '26, the 7th ACM International Conference on AI in Finance (Milan, Nov 14–17, 2026). All figures are as printed in the submitted paper.

What changed in the call, not who is talking

A one-minute read for someone new to the topic. Slides 03–09 give the evidence.

In plain terms

The problem Earnings-call Q&A holds timely information, but most of its language is what that company always says. A model trained on raw transcripts learns to recognise the company rather than what is new.

The idea Turn each call into a vector, subtract the same firm's own past-call average and keep the signed direction of the change. Average the changes by sector and rank the 11 US sector ETFs each month.

The result Over 24 out-of-sample months (Dec 2023–Apr 2026) the monthly sector-rank IC was .158; buying the top-ranked sector returned 42.1% a year after costs versus 24.0% for SPY, with about half the drawdown.

MY ROLE · WHERE THIS COULD BE USED

My role Third author.

Use case A monthly sector-rotation signal from transcripts, and a general recipe for removing identity from text embeddings.

Terms used in this deck

EMBEDDING

A vector of numbers that represents a passage of text; here from Industry-BERT-SEC, a finance-tuned language model.

SIGNED DISPLACEMENT

The vector difference between this call and the firm's own past calls; it keeps direction, unlike a distance.

IC (INFORMATION COEFFICIENT)

Correlation between the predicted sector ranking and the realised one; 0 is a coin flip.

SECTOR ETF

A fund tracking one US sector, such as Technology; SPY tracks the whole S&P 500.

OUT OF SAMPLE

Decisions made on months the model never saw in training; the encoder was released before the test began.

NEWBY-WEST p-VALUE

A significance test that allows for overlapping, autocorrelated monthly returns.

Isolate what changed, not who is talking

Earnings-call Q&A is timely, but most of its language repeats what the same company always says.

Why it matters

- Sector indicators confirm shifts after they begin. The less-scripted Q&A adds information beyond the announcement.
- Calls from the same company cluster in embedding space, so a model can learn the issuer instead of what is new.
- Distance throws away direction: stronger demand and rising cost pressure can sit at the same distance from past language.
- Pooling firms into sectors lets large issuers and long transcripts dominate.

WHAT THIS MEANS OUTSIDE THE PAPER

Each firm vs its own past what is left is what changed, not who is speaking.

Frozen before the test heads fit through 2020, then unchanged for 24 decisions.

Generalizes to any embedding where identity dominates the signal you want.

Data & setup

TRANSCRIPTS

Financial Modeling Prep; 22,897 parsed → 21,421 locked calls

MARKETS

11 U.S. sector ETFs vs SPY; Norgate constituents; Yahoo Finance prices

OUT OF SAMPLE

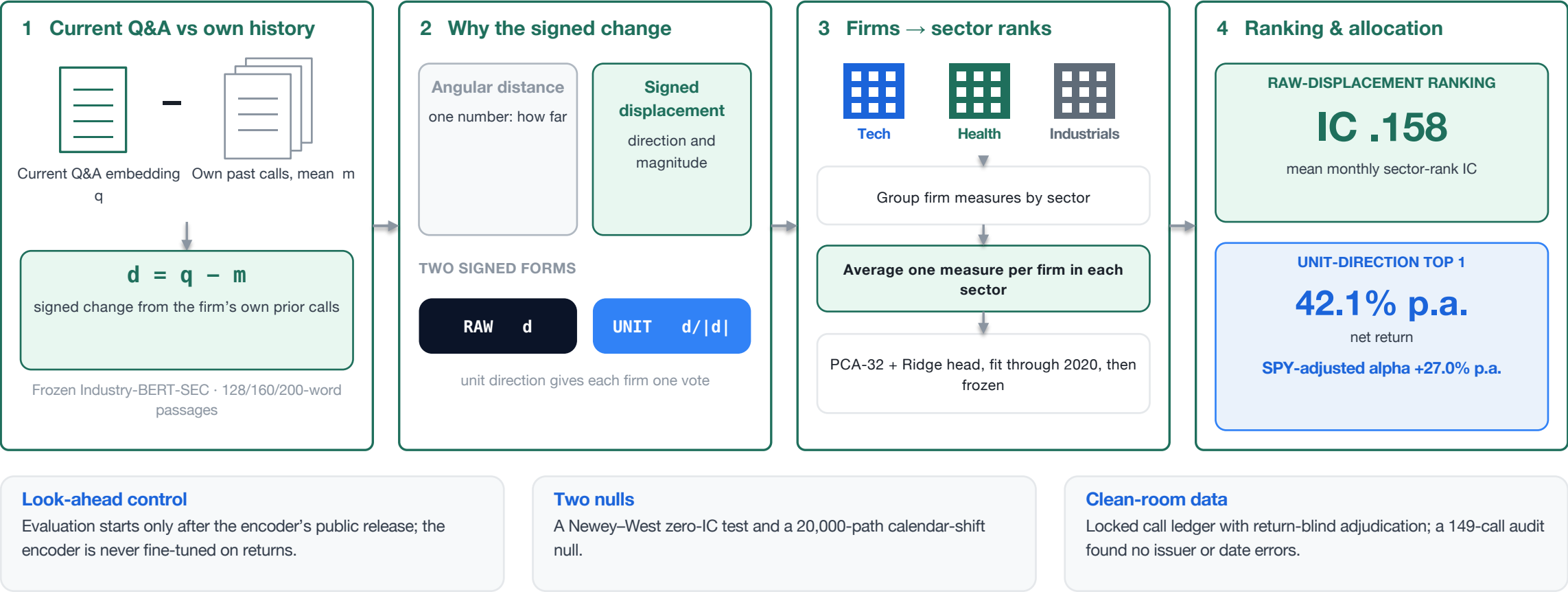
24 monthly decisions, Dec 2023–Apr 2026, after the encoder's release

TRAINING

Ranking heads fit only on outcomes ending by Dec 1, 2020, then frozen

From a firm's Q&A change to a sector bet

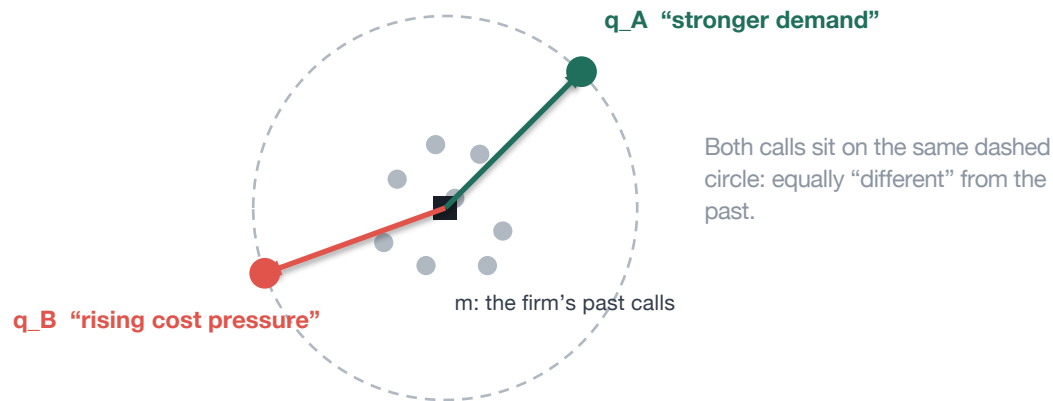
Each call is compared with the same firm's past calls; the signed changes are averaged by sector and ranked.



Why the signed change, and why each firm's own history

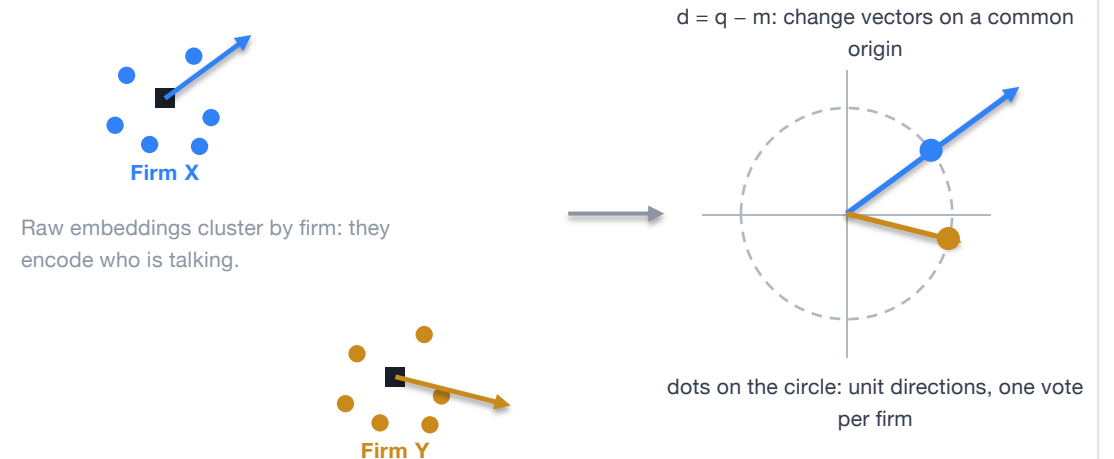
Distance alone cannot tell opposite news apart, and raw embeddings mostly encode who is speaking.

Distance hides direction



A distance feature scores both calls the same. Signed displacement keeps the direction, so the two calls send opposite signals.

Subtract each firm's own history



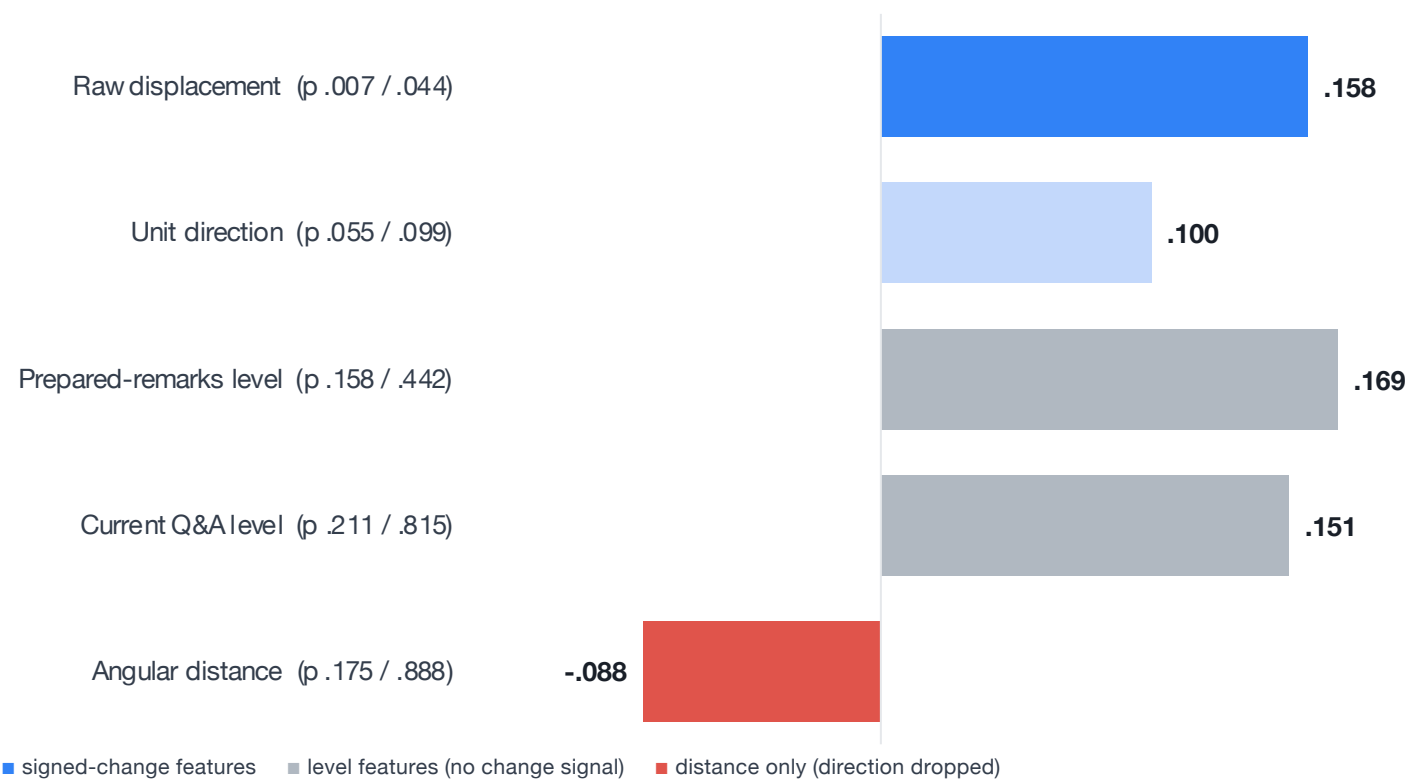
Subtracting the firm's own mean removes the part of the embedding that just identifies the firm; what remains is what changed.

Illustration of the idea in the paper's §3 and Eqs. 1–5; the points are schematic, not data.

The signed change carries the signal, not the company name

Two level features score as high as the winner and both fail the tests. Only raw displacement survives both. p-values are Newey–West / calendar-shift.

Sector-rank IC



Source: CallRank Tables 2–3 and Fig. 2.

What it shows

- IC is the correlation between the predicted ranking and what actually happened. 0 is a coin flip; .158 monthly is a strong sector signal.
- Raw displacement is the only representation that passes both nulls. Level features with a similar IC fail both.
- Angular distance, which drops direction, scores $-.088$.
- Conditioning on each firm’s own history cuts issuer retrieval from 41.4% to 3.2%: the vector carries the change, not the name.
- Separately, masking company, ticker and speaker names from the text still leaves IC .191, while a body-swap placebo falls to $-.057$.

After costs, the top-ranked sector beats SPY

Top 1 allocation (unit direction), net of 5 bp one-way costs; decisions Dec 2023–Apr 2026.

Portfolio	Ann. net return	Sharpe	Max drawdown	Beta
CallRank Top 1	42.1%	2.05	9.9%	0.55
SPY	24.0%	1.17	18.8%	1.00
LM Q&A tone residual	20.1%	0.95	17.6%	0.62
1/N sectors	17.2%	0.96	15.3%	0.73
CallRank Bottom 1	13.0%	0.57	24.4%	0.77
42-day momentum	11.6%	0.47	20.3%	0.66
63-day low volatility	7.3%	0.27	11.1%	0.34

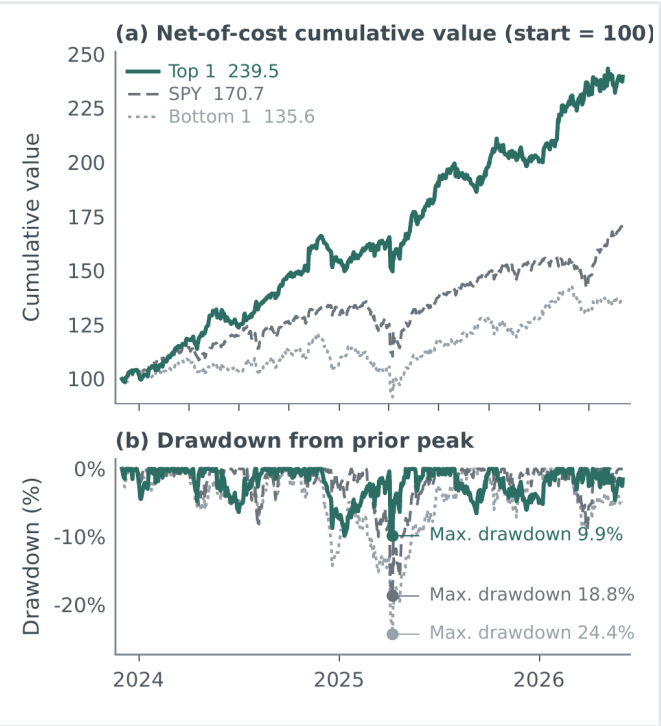
What it shows

- Alpha: 27.0% SPY-adjusted, 24.0% CAPM, 20.3% Fama–French 5 + momentum.
- At 20 bp one-way costs: still 39.6% net and 24.8% SPY-adjusted alpha.

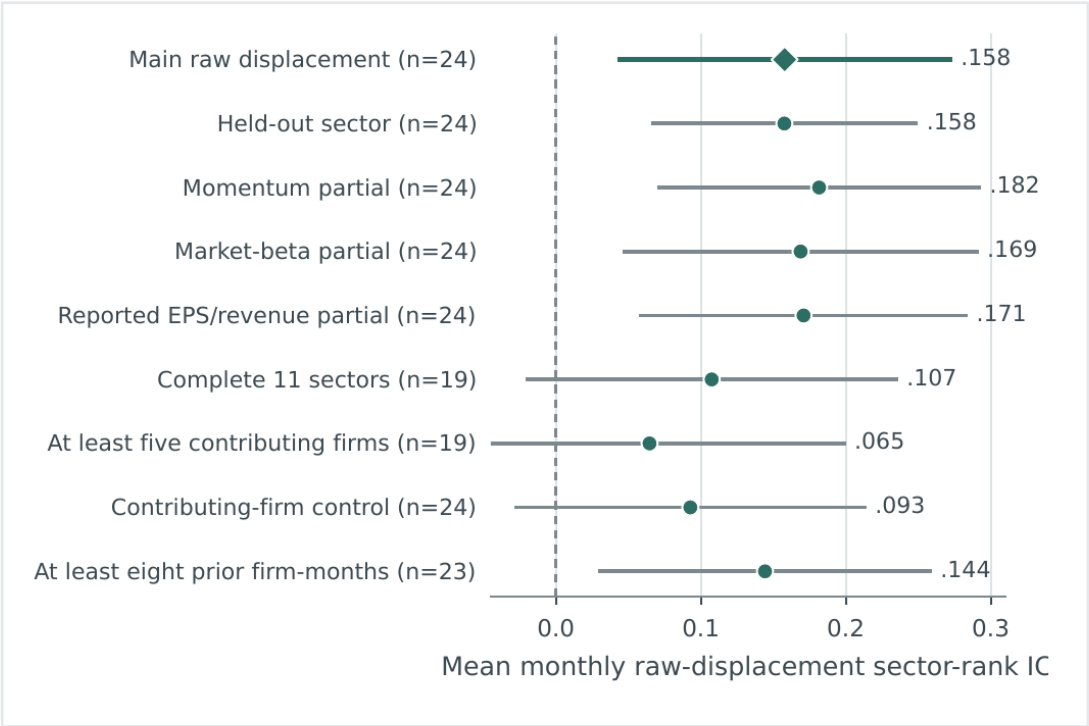
Source: CallRank Table 4, Fig. 3 and §5.4.

The return path and robustness, as published

Original figures from the submitted paper, shown as published. Numbers on earlier slides are read from its tables and text.



Paper Figure 3: Net-of-cost value and drawdown, Top 1
Growth of 100: 239.5 for Top 1 vs 170.7 for SPY; drawdown 9.9% vs 18.8%.



Paper Figure 4: Robustness of raw-displacement sector-rank IC
All nine point estimates are positive; six of nine 95% Newey–West intervals exclude zero.

What held, and what the paper does not claim

The two headline numbers come from two different representations; the paper says so explicitly.

Checks that held

- Against 2,000 random sector paths with the same accounting, CallRank sits in the favorable tail for return, Sharpe, alpha and drawdown (all $p \leq .009$).
- Momentum, market-beta and reported-earnings partial ICs stay at .169–.182.
- Picks span 10 sectors and change in 16 of 23 transitions; Top 1 beats Bottom 1 in every calendar year.
- Post hoc: replacing Utilities when it ranks first still gives 29.3% net.
- Industry-BERT-SEC is the only matched encoder that rejects both nulls at a common passage length.

What the paper does not claim

- The IC headline (.158) uses raw displacement. The portfolio uses unit direction, whose own IC is .100 (NW $p .055$). The paper claims neither dominates.
- It predicts which sector leads, not why, and not individual stocks.
- Only 24 monthly decisions; borrow costs excluded.
- The encoder's pretraining cutoff is undisclosed; starting evaluation after its release is the control.

Code (anonymous review repository): anonymous.4open.science/r/callrank-icaif26-v11-review-artifact-740C